Chapter 1: Vectors and Functions

October 8, 2025

Outline

What is a vector?

- A vector in $\mathbb{R}^n$ is an ordered n-tuple $\mathbf{x} = (x_1, \ldots, x_n)$.
- Think: directed segment from origin to the point with coordinates x_i.
- Vector space operations (component-wise addition and scalar multiplication).

Basic operations

Addition: $\mathbf{x} + \mathbf{y} = (x_1 + y_1, \dots, x_n + y_n)$.
Scalar multiplication: $\alpha\mathbf{x} = (\alpha x_1, \dots, \alpha x_n)$. **Properties:** associativity, commutativity of addition, distributivity. **Example:** For $\mathbf{x} = (1, 2, 3),\ y = (0, -1, 2)$, compute $2\mathbf{x} - 3\mathbf{y} = (2 - 0, 4 + 3, 6 - 6) = (2, 7, 0)$.

Dot product and inner product

Definition: For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^{n} x_i y_i .$$

Properties:

- Symmetry: $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$.
- Linearity in each slot.
- Positive-definiteness: $\mathbf{x} \cdot \mathbf{x} \geq 0$, equality iff $\mathbf{x} = 0$.

Norm and distance

Norm: $\|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}}$. **Distance:** $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$. **Cauchy–Schwarz Inequality:**

$$|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \, \|\mathbf{y}\|.$$

Triangle inequality:

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|.$$

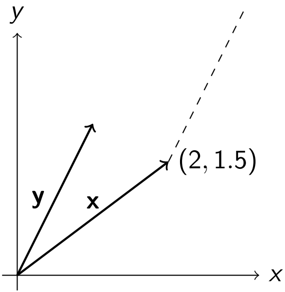
Proof sketch: Cauchy–Schwarz

Consider the quadratic polynomial in t:

$$0 \leq \|\mathbf{x} + t\mathbf{y}\|^2 = (\mathbf{x} + t\mathbf{y}) \cdot (\mathbf{x} + t\mathbf{y}) = \|\mathbf{x}\|^2 + 2t(\mathbf{x} \cdot \mathbf{y}) + t^2\|\mathbf{y}\|^2.$$

Discriminant ≤ 0 gives $(\mathbf{x} \cdot \mathbf{y})^2 \leq \|\mathbf{x}\|^2\|\mathbf{y}\|^2$.

Geometry in $\mathbb{R}^2$ and $\mathbb{R}^3$



Visualize projection, angle: $\cos\theta = \dfrac{\mathbf{x}\cdot\mathbf{y}}{\|\mathbf{x}\|\,\|\mathbf{y}\|}$.

Functions $f : \mathbb{R}^n \to \mathbb{R}^m$

- A function f assigns to each $\mathbf{x} \in \mathbb{R}^n$ a vector $f(\mathbf{x}) \in \mathbb{R}^m$.
- Component form: $f(\mathbf{x}) = (f_1(\mathbf{x}), \ldots, f_m(\mathbf{x}))$ where each $f_i : \mathbb{R}^n \to \mathbb{R}$.

Examples:

- Linear map: $A : \mathbb{R}^n \to \mathbb{R}^m$, $A(\mathbf{x}) = M\mathbf{x}$ for an $m \times n$ matrix M.
- Polynomial maps, coordinate projections, norms, etc.

Limits of functions

Definition: $\lim_{x \to a} f(\mathbf{x}) = b$ means: for every $\varepsilon > 0$, there exists $\delta > 0$ s.t. $\|\mathbf{x} - a\| < \delta, x \neq a$ implies $\|f(\mathbf{x}) - b\| < \varepsilon$.

Note: Use Euclidean norm in the domain and the co-domain.

Continuity

f is continuous at a iff $\lim_{x \to a} f(x) = f((a)$.
Sequential characterization: f is continuous at a iff whenever $x_k \to a$, we have $f(x_k) \to f(a)$.

Examples:

- Polynomials, linear maps are continuous everywhere.
- Norm function $\mathbf{x} \mapsto \|\mathbf{x}\|$ is continuous.

Operations preserve continuity

If $f, g : \mathbb{R}^n \to \mathbb{R}^m$ continuous and $\alpha : \mathbb{R}^n \to \mathbb{R}$ continuous, then

- $f + g$ and αf are continuous (component-wise).
- Composition: if $g : \mathbb{R}^n \to \mathbb{R}^m$ continuous and $h : \mathbb{R}^m \to \mathbb{R}^p$ continuous, then $h \circ g$ continuous.

Examples: Limits that fail

Consider $f : \mathbb{R}^2 \setminus \{0\} \to \mathbb{R}$, $f(x, y) = \dfrac{xy}{x^2 + y^2}$.

- Along $y = x$ we have $f = \dfrac{x^2}{2x^2} = \frac{1}{2}$ (as $x \to 0$).
- Along $y = 0$ we get 0.

Thus $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist.

Directional limits and continuity

Directional limit along unit vector u: $\lim_{t \to 0^+} f(a + tu)$. If directional limits differ for different u, an ordinary limit does not exist.

Frechet derivative (brief intro)

A function $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at $ifthereexistsalinearmap L: R^n \to \mathbb{R}^m$ such that

$$\lim_{\to 0} \frac{\|f() - f((a) - L()\|}{\|\|} = 0.$$

In that case L is the derivative (Jacobian matrix).

Jacobian matrix

If $f = (f_1, \ldots, f_m)$ and partial derivatives $\partial f_i / \partial x_j$ exist and are continuous, then

$$Df((a) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}((a) & \cdots & \frac{\partial f_1}{\partial x_n}((a) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}((a) & \cdots & \frac{\partial f_m}{\partial x_n}((a) \end{pmatrix}.$$

This matrix represents the best linear approximation near .

Worked example 1: Limit

Show $\lim_{(x,y)\to(0,0)} \dfrac{x^3}{x^2+y^2} = 0$.

Solution sketch.

Use polar coordinates: $|x| \le r$, so

$$\left|\frac{x^3}{x^2+y^2}\right| = \frac{|x|^3}{r^2} \le \frac{r^3}{r^2} = r \to 0 \ (r \to 0).$$

$\square$

Worked example 2: Continuity

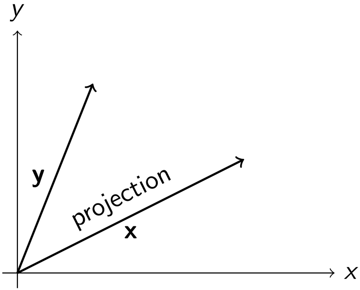
Show $f(\mathbf{x}) = \dfrac{\mathbf{x}}{1 + \|\mathbf{x}\|}$ is continuous on $\mathbb{R}^n$.

- Composition of continuous maps: $\mathbf{x} \mapsto \|\mathbf{x}\|$ then $1 + \|\mathbf{x}\|$ then reciprocal then product.

Exercise set

1. Prove triangle inequality directly from dot product properties.
2. Determine whether $g(x, y) = \dfrac{x^2 - y^2}{x^2 + y^2}$ has a limit at $(0, 0)$.
3. Compute the Jacobian of $f(x, y) = (x^2 y, e^{xy})$ and verify differentiability at $(0, 0)$.

Vector projections (figure)



Polar / Cylindrical view (useful trick)

For $\mathbb{R}^2$, use polar: $x = r\cos\theta, y = r\sin\theta$ to analyze radial behavior.
For functions with homogeneous degrees, polar often simplifies limits.

References and further reading

- S. R. Ghorpade & B. V. Limaye, *A Course in Calculus and Real Analysis*, Chapter 1.
- Standard multivariable calculus texts for more geometric intuition.

What next?

- I can split this into smaller presentations for each major subsection (Vectors, Dot product, Functions, Limits, Differentiability).
- I can produce a printable handout (PDF) or annotate slides with solutions.
- I can add more TikZ figures or animation frames.